

Experiment 18: Perceptron-based IRIS Classification

Aim

To implement **Perceptron-based Iris classification** using Python.

Algorithm

Step 1: Import the required libraries.

Step 2: Load the Iris dataset.

Step 3: Select the input features and target classes.

Step 4: Split the dataset into training and testing sets.

Step 5: Create a Perceptron classifier.

Step 6: Train the model using the training data.

Step 7: Predict the classes for the test data.

Step 8: Calculate and display the accuracy.

Program

```
#Experiment 18: Perceptron-based IRIS Classification
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import Perceptron
from sklearn.metrics import accuracy_score

iris = load_iris()

X = iris.data
y = iris.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = Perceptron(random_state=42)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Predictions:", y_pred)
print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
ndarray: y_test
ndarray with shape (30,)
```

Output

```
... Predictions: [1 0 1 1 1 0 1 1 1 1 0 0 0 0 1 1 1 1 1 0 1 0 1 1 1 1 1 0 0]
Accuracy: 0.6333333333333333
```

Result

Thus, the **Perceptron-based Iris classification was successfully implemented**, and the Iris flower classes were predicted using the trained Perceptron model.